Macro-Systematic Approach to Multi-Asset Investing

Risk Profile Analysis: Commodities (SXRS)

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Alpha Capital Jr.
Buy-Side

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Executive summary

This document is part of a macro-driven multi-asset allocation research project. The objective is to design a rules-based framework that translates macroeconomic regimes into systematic portfolio allocations and decisions across a set of assets, targeting risk-adjusted alpha over equity benchmarks.

This section covers individual asset analytics: return distribution, performance, risk, volatility, Sharpe, Sortino, VaR, CVaR, maximum drawdown, and time under water. Each asset is analyzed independently before being integrated into the broader cross-asset and allocation framework.

This report focuses on Commodities (SXRS), through an ETF containing Oil, metals, corn etc...

Assets choice

Class	Product	Ticker	First Obs.
Equity	ETF S&P 500	SPY	29-01-1993
Commodities	SPDR Gold Shares	GLD	18-11-2004
Commodities	iShares Diversified Commodity Swap	SXRS.DE*	02-01-1991
Bonds	Treasuries 20Y+	TLT	30-07-2002
Credit	iShares Corp Bond	LQD	TBD

Table: Selected assets for a multi-asset portfolio

* I previously planned to use DBC, but SXRS is in fact a better commodity exposure instrument.

Basket of commodities - General Description

Characteristics:

- **Underlying:** Bloomberg Commodity Index Total Return
  - Single asset class
  - Holdings various commodities
- **Synthetic** replication via Swaps.
- **Cheap & highly liquid** ETF
  - Volume
  - $2B AUM
  - 0.19% expense ratio
- Accessible in Europe (UCITS & KID)

→ This asset is a **Tactical element** for "Policy Tight" scenarios, with no long-term holding purposes.

Commodities - Composition

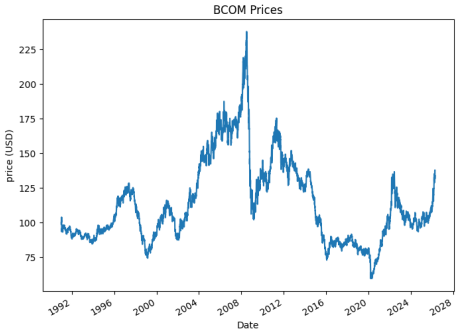
Composition of the Bloomberg Commodity Index :

Sector	**commodities**	**Weights**
Energy	(WTI, Brent, RBOB, ULSD)	30%
Agriculture	(Corn, Soy, Wheat, Sugar, Cotton, etc.)	30%
Metals	(Gold, Silver, Copper, Aluminum)	30%
Livestock	(Live Cattle, Lean Hogs)	10%

Table: Composition Bloomberg Commodity Index

Commodities - Historic performances

The analysis uses the **Bloomberg Commodity Index**, with data available since 1991 (vs. 2018 for SXRS.DE). Results should be interpreted cautiously, as ETF pricing dynamics differ (Fees, adjustments, structural frictions)



Performance:

- **Period:** 1991–2026 (35 years)
- **CAGR:** 0.83%
- **Annualized Volatility:** 14.87%
- **Sharpe Ratio:** 0.026
- **Sortino Ratio:** 0.036

Insight: A CAGR of 0.83% for the BCOM index confirms commodities should not be held long-term. However, its 14.87% volatility creates tactical opportunities in specific macro regimes.
Further analysis identifies the macroeconomic regimes in which commodities are most effective.

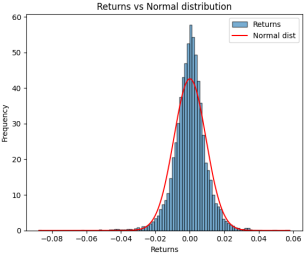
Commodities - Distribution of returns

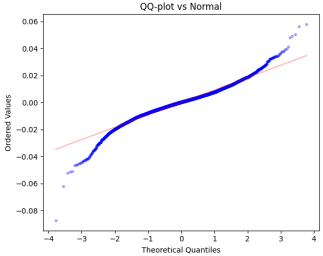
Distribution parameters

Metric	Value
Mean	0.0077%
Skewness	-0.3273
Kurtosis	3.9622
Exc. Kurtosis	0.9622

Risk metrics

CL	VaR	CVaR
95%	1.49%	2.24%
97.5%	1.93%	2.80%
99%	2.66%	3.66%





- **Left-tail bias** from negative skew
- Excess kurtosis is modest, suggesting relatively **Thin tails**
- **QQ-plot** near-linear: minimal deviation from Gaussian at both tails; Normal VaR appears to approximate tail risk reasonably well.

→ *Insights:* BCOM offers tactical volatility exposure without SPY/GLD's black-swan tail risk, Gaussian approximation is safe for rapid position sizing in inflation/policy-tight regimes; enter on spike, exit post reversion.

Commodities - Mean reversion

Is the Bloomberg commodity index Mean-reverting ?:

→ As BCOM's flat long-term CAGR, high volatility, and sideways price behavior initially suggest a mean-reverting profile.

ADF test results:	
Test Statistic	-1.916
p-value	0.325
Critical Value (5%)	-2.862

KPSS test results:	
Test Statistic	2.163
p-value	0.010
Critical Value (5%)	0.463

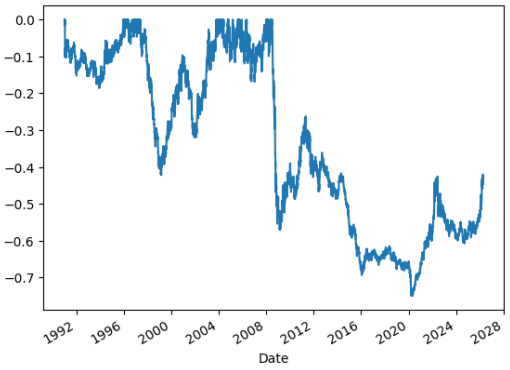
→ Both tests reject a mean-reverting specification.

- ADF fails to reject the unit root.
- KPSS rejects stationarity.

Insight: While there are no mean-reverting properties, this asset exhibits a slow trend with local cycles; Allocation should therefore rely on regime timing and momentum.

Commodities - Drawdowns analysis

Historical drawdowns:



Important dates:

- **2008 – Commodity supercycle crash**
- **2020 – COVID-19** Oil & Energy demand massively drops.
- **2022 – Russia-Ukraine War** Shock accross all commodity sectors..

Observations

- Major shocks created drawdowns that remained underwater for extended periods.
- Smaller drawdowns, typically around -10%, tend to recover within roughly 100 days.

Insights: The drawdown analysis confirms that commodities show strong and rapid reversals. Risk lies in long exposures, not in daily volatility.

Commodities - Drawdowns Tables

No major shock (2006)
Rapid correction, rapid recovery

Metric	Value
Peak	May 13, 1996
Trough	Jul 26, 1996
Recovery	Oct 8, 1996
Max Drawdown	-6.59%
TUW	103d
Recovery Time	51d

High Volatility period
Ranged between [75$ and 125$]

Metric	Value
Peak	May 16, 1997
Trough	Feb 26, 1999
Recovery	Nov 13, 2003
Max Drawdown	-42.22%
TUW	1628
Recovery Time	1182d

High volatility momentum during
Third Oil Shock (2003-2008)

Metric	Value
Peak	Oct 26, 2004
Trough	Dec 10, 2004
Recovery	Mar 8, 2005
Max Drawdown	-11.32%
TUW	91d
Recovery Time	59d

Financial Crisis
Hit ATH: 225$, economy slows down: 110$, rebound: 175$

Metric	Value
Peak	May 11, 2006
Trough	Jan 9, 2007
Recovery	Jan 2, 2008
Max Drawdown	-16.88%
TUW	413d
Recovery Time	247d

Another High volatility momentum during
Third Oil Shock (2003-2008)

Metric	Value
Peak	Dec 13, 2005
Trough	Mar 8, 2006
Recovery	May 9, 2006
Max Drawdown	-11.91%
TUW	100d
Recovery Time	43d

Notes:
This tables confirm my assumptions about repeated sharp drawdowns and recoveries. It tends to remain stable for long periods before external shocks trigger abrupt moves that are often followed by quick reversion and renewed stability.
Commodities require further analysis because their behavior is strongly regime-dependent.

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Macro-Systematic Approach to multi-asset investing

Nicolas Poncin is a M.Sc. Financial Engineering student with a strong interest in global macroeconomics and multi-asset portfolio management. He aims to apply his academic background to real investment decision-making. This project was developed as a practical research initiative to explore how macroeconomic regimes influence asset behaviour and portfolio allocation, intending to build a disciplined and testable investment process.



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